

A PRELIMINARY REPORT ON

**Innovative Deep Learning Approach for Parkinson's Disease
Prediction: Leveraging Convolutional Neural Networks for Early
Detection**

SUBMITTED IN THE PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE AWARD OF THE DEGREE

OF

**BACHELOR OF TECHNOLOGY
(Artificial Intelligence)**

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2023-24



CERTIFICATE

This is to certify that the project report entitles

**“INNOVATIVE DEEP LEARNING APPROACH FOR PARKINSON’S
DISEASE PREDICTION: LEVERAGING CONVOLUTIONAL NEURAL
NETWORKS FOR EARLY DETECTION”**

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ABSTRACT

Parkinson's disease (PD) is a progressive neurodegenerative disorder that primarily affects movement control. Early diagnosis and intervention are critical for improving patient outcomes and quality of life. PD poses a significant challenge for accurate diagnosis due to its similarities with other neurological conditions, resulting in a 25% rate of inaccurate manual diagnoses. Convolutional neural networks (CNN) are a type of deep learning model that have been shown to be effective in image analysis as they are able to learn complex patterns in images and identify subtle changes that may be indicative of disease. In this project, we introduce an innovative CNN-based automated diagnostic system that effectively distinguishes PD from healthy controls (HC). We have used the Parkinson Progression Markers Initiative (PPMI) dataset, which provides publicly available benchmarked magnetic resonance imaging (MRI) images for both PD and healthy controls (HC). It contains a total of 31436 records of MRI scans and is categorized into PD (18690) and healthy control (12746). Furthermore, data is divided into training, validation, and testing sets with a ratio of 80%, 10%, and 10%, respectively.

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1. INTRODUCTION

1.1 OVERVIEW

Parkinson's Disease (PD) is a neurodegenerative disorder first described by James Parkinson. In Parkinson's Disease, several nerve cells (neurons) in the brain gradually break down or die. The gradual decrease in dopamine level causes abnormal brain activity, leading to impaired movement and other symptoms of PD. It has been reported by the Centers for Disease Control and Prevention (CDC) that PD complications rank as the 14th leading cause of death in the United States (US). Till now, there is no way to cure PD permanently. Meditation and proper medical treatment decreases the progress of PD. Therefore, detection of PD at an earlier stage is certainly important to slow down its progression.

The signs and symptoms can be different for everyone. Early signs may be mild and unnoticed. The signs and symptoms may include tremors, slow body movement, (bradykinesia), rigid muscles, imbalance, loss of automatic movements, speech changes, or writing changes.

Thus, we have developed fast and efficient predictive model to detect PD. The main aim was to develop CNN model which will efficiently classify PD affected MRI images and HC images.

1.2 MOTIVATION OF THE PROJECT

Early diagnosis of PD is challenging because the symptoms are often subtle in the early stages of the disease. This can lead to delays in diagnosis, which can have a negative impact on quality of life and the progression of the disease. Deep learning has the potential to overcome the challenges of early diagnosis of PD. It can learn complex patterns in data that would be difficult to identify with traditional machine learning algorithms. This means that deep learning algorithms could be used to identify early signs of PD that are not currently detectable with other methods.

In short, the motivation for using deep learning to predict PD is to improve the accuracy and timeliness of diagnosis. The disease is characterized by a loss of dopamine-producing neurons in the brain. There is no cure for PD, but early diagnosis and treatment can help to

slow the progression of the disease and improve quality of life. Currently, the diagnosis of PD is based on a combination of clinical symptoms, imaging tests, and genetic testing. However, these methods are not always accurate, and early diagnosis can be difficult. Convolutional neural networks are a type of deep learning algorithm that are specifically designed for image processing. CNNs have been shown to be very effective in a variety of tasks, including image classification, object detection, and face recognition.

In recent years, there has been a growing interest in using CNNs for the early detection of PD. CNNs can be used to analyze video recordings of patients' movements, and they can be trained to identify subtle changes in motor function that are characteristic of PD. A number of studies have shown that CNNs can be used to achieve high accuracy in the early detection of PD. For example, one study found that a CNN was able to correctly identify PD patients with 90 to 92% accuracy. The early detection of PD is important for a number of reasons. First, early diagnosis can allow patients to start treatment earlier, which can help to slow the progression of the disease. Second, early diagnosis can help patients to access clinical trials for new treatments. Third, early diagnosis can help patients to make informed decisions about their future care. The rapid advancements in computer vision and deep learning have led to breakthroughs in various fields, including medical imaging analysis. By applying these cutting-edge technologies to Parkinson's disease detection, we have an opportunity to explore new insights and discover potential biomarkers related to the disease.

Implementing a successful deep learning model for Parkinson's disease detection reinforces the potential of artificial intelligence (AI) in transforming healthcare. By showcasing AI's applicability in real-world medical scenarios, this research can inspire further investment and interest in developing AI-based solutions for other medical challenges. Given the potential benefits of early detection, there is a strong motivation for developing and deploying CNN-based methods for the early detection of PD.

1.3 PROBLEM DEFINATION

To develop the predictive model which distinguishes Parkinson's Disease patients from healthy group using Convolutional Neural Network.

1.4 GOALS AND OBJECTIVES

- To convert DCM extension MRI images to jpeg.jpg or png image format
- To resize images for fixed width and height
- To remove the noise and do area filtration in data
- To train the CNN model using MRI images

1.5 PROJECT SCOPE

One key future scope is to extend the model's capabilities by incorporating multi-modal data, including not only brain MRI images but also patient clinical data, genetic information, and vocal features. By integrating these diverse data sources, the model can provide a more comprehensive and accurate diagnosis, potentially revolutionizing early PD detection.

Continuous monitoring of disease progression is an intriguing future scope. Extending the model's capabilities to track PD progression over time can assist in personalized treatment planning and optimize patient care.

2. LITERATURE SURVEY

In recent years, there has been a growing interest in the early detection of Parkinson's disease (PD) using advanced machine learning techniques. Several studies have explored the use of Convolutional Neural Networks (CNNs) for this purpose [1]. In a 2018 study published on ResearchGate, researchers developed a customized CAD-based CNN architecture to classify MRI patches of both Parkinson's and healthy brain patterns [13]. Despite achieving an impressive accuracy of 96%, their progress was hindered by the limitation of a small dataset, which led to concerns of overfitting. In a study, titled "Detection of Parkinson Disease in Brain MRI using Convolutional Neural Network," a customized CNN architecture based on Computer-Aided Diagnosis (CAD) was proposed [2]. This approach demonstrated promise in classifying MRI patches indicative of Parkinson's disease and healthy patterns [5]. However, the study faced challenges associated with a limited dataset, which led to concerns of overfitting, despite achieving an accuracy rate of 96%.

Moving forward to 2020, another study titled "Automatic Parkinson's Disease Detection at Early Stages as a Pre-diagnosis Tool by Using Classifiers and a Small Set of Vocal Features" explored the use of classifiers such as k-nearest neighbor, multi-layer perceptron, support vector machine, and random forest for early-stage Parkinson's disease detection [9]. This research leveraged vocal features as potential biomarkers. Nevertheless, the analysis was constrained by background noise in speech recordings, resulting in a notable number of false alarms and missed detections, albeit achieving an accuracy of 94.7% [3].

In a 2021 IEEE publication, the focus shifted towards a broader range of machine learning approaches, including ensemble methods like Boosting, Random Forest, Logistic Regression, Discriminant Analysis, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Supervised Feed Forward Networks [7][8]. While their target accuracy of 96.45% was attained, it was noted that achieving such accuracy levels required a substantial amount of training data, which can be a challenging limitation in practice [4].

A study in May 2022 in the National Library of Medicine introduced a fuzzy ensemble model named FRLF, applied to the confidence scores of four classic deep learning models (VGG16,

ResNet50, Inception-V3, and Xception) for the detection of Parkinson's disease using DaTscan images. Remarkably, this approach achieved an accuracy of 98.45% [5]. However, a noteworthy challenge in this study was the unbalanced dataset, with a higher number of PD patients than healthy controls, potentially leading to overfitting issues specific to the PD patients.

In a more recent study from January 2023 titled "Early Detection of Parkinson's Disease using Machine Learning," the focus shifted to the classification of Parkinson's disease using vowel phonation data. This research achieved a noteworthy 91.835% accuracy with a Random Forest classifier, along with a sensitivity of 0.95 [6]. Moreover, the study highlighted the results of an SVM model, which attained an accuracy of 91.836% and a sensitivity of 0.94 after applying Principal Component Analysis (PCA) to the dataset. These models were particularly robust in handling outliers [11][18]. However, it's essential to acknowledge that the project solely relied on vowel phonation data, potentially lacking a comprehensive biomarker for Parkinson's disease classification.

Collectively, these studies underscore the progress made in leveraging advanced technologies like CNNs and machine learning algorithms, along with vocal features, for the detection of Parkinson's disease. While each approach exhibits strengths, the limitation of datasets and the need for further exploration of comprehensive biomarkers remain prominent challenges in this evolving field of research.

Table 1: Literature Survey

TITLE	ALGORITHMS USED	LIMITATIONS
Predicting the occurrence of Parkinson's Disease using various classification models (2018) Journal: ResearchGate	The model used voice dataset consists of voice frequencies. Various machine learning algorithms were applied on the dataset and among them • ANN has shown highest accuracy (94.87%). • Random Forest which is a Classification algorithm has shown good accuracy (87.17%) while • Naive Bayes has shown least accuracy (71.79%). Accuracy – 94.87 %	The models were not able to achieve better balance between sensitivity and specificity.
Detection of Parkinson Disease in Brain MRI using Convolutional Neural Network (2018) Journal: ResearchGate	A customized CAD based CNN architecture can be used to classify MRI patches of Parkinson and healthy patterns. Accuracy – 96%	Limited dataset was major issue, leading the CNN model towards overfitting.
Automatic Parkinson's disease detection at early stages as a pre-diagnosis tool by using classifiers and a small set of vocal features (2020) Journal:	• k nearest neighbor, • multi-layer perceptron, • support vector machine • random forest. Accuracy – 94.7%	The performance of PD detection based on speech analysis is generally limited by background noise in speech recording, causing a high number of false alarms and missed detection.

ResearchGate		
Early Detection of Parkinson's Disease Using Deep Learning and Machine Learning (2021) Journal: IEEE	• Ensemble method – Boosting algorithm, • Random forest, • Logistic Regression, • Discriminant Analysis, • KNN, • Support vector machine • Supervised feed forward network. Accuracy – 96.45%	Targeted accuracy was not achieved as model needs large amount of data for training purpose.
The parkinson's disease detection using machine learning techniques (2021) Journal: IJRET	• XGBoost, • Decision Tree Classifier, • Naive Bayes. Testing - Unit testing, Integration testing, Functional testing. Accuracy – XGBoost - 94% Decision Tree Classifier - 87% Naive Bayes - 75%	The model used a small dataset of voice recordings, which limits the generalizability of the findings. It did not evaluate the robustness of the model to noise and other factors that could affect the voice signal.
An Ensemble of CNN Models for Parkinson's Disease Detection Using DaTscan Images (2022) Journal: National Library of Medicine	The model predicts Parkinson's disease effectively using the PPMI DaTscan images. It is a fuzzy ensemble model, called FRLF, which is applied on the confidence scores of four classic DL models- VGG16, ResNet50, Inception-V3, and Xception to enhance the overall results of the model. Accuracy – 98.45% (CNN model + FRLF)	• The dataset is unbalanced, with more PD patients than healthy controls. This can lead to overfitting of the models to the PD patients. • The project only used voice samples from a single recording session. This may not be enough to capture the full range of vocal

		changes that can occur in PD.
Early detection of Parkinson's disease using machine learning (2023) Journal: ScienceDirect	Parkinson's disease classification using vowel phonation data gives an 91.835% accuracy and 0.95 sensitivity for Random Forest classifier. Results of the Random Forest model are ideal, due to equal importance given to all 22 attributes in MDVP dataset. This paper also highlights the results of the SVM model that gives an accuracy of 91.836% and sensitivity of 0.94, after PCA is applied to the dataset. Both SVM and Random Forest models perform well for outliers and are robust models. Accuracy – SVM - 91.836%	The project only used vowel phonation data, which may not be a sufficient biomarker for classifying Parkinson's disease.
Machine learning approaches to identify Parkinson's disease using voice signal features (2023) Journal: National Library of Medicine	The model detect PD using different types of Machine Learning (ML), and Deep Learning (DL) models such as – • Support Vector Machine (SVM), • Random Forest (RF), • Decision Tree (DT), • K-Nearest Neighbor (KNN), and • MultiLayer Perceptron (MLP) to differentiate between healthy and PD patients by voice signal features. Accuracy – SVM - 95% MLP - 98.31%	• The dataset is unbalanced, with more PD patients than healthy controls. This can lead to overfitting of the models to the PD patients. • The project only used voice samples from a single recording session. This may not be enough to capture the full range of vocal changes that can occur in PD.

3. SYSTEM REQUIREMENTS

3.1 HARDWARE REQUIREMENTS

- RAM: 8.0 GB/16.0 GB
- System Type: 64-bit operating system, x64-based processor
- Processor: 11th Gen Intel(R) Core (TM) i5-11260H @ 2.60GHz 2.61 GHz

3.2 SOFTWARE REQUIREMENTS

- Operating System (OS): Windows 7 or higher version
- Integrated Development Environment (IDE): Visual Studio Code
- Programming Language: Python, CSS, HTML, Javascript
- Database: phpMyAdmin (MySQL)
- Libraries: Keras, NumPy, CV2, sklearn, Matplotlib, SciPy, TensorFlow, Flask
- Web Browser: Compatible with any web browser

4. SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

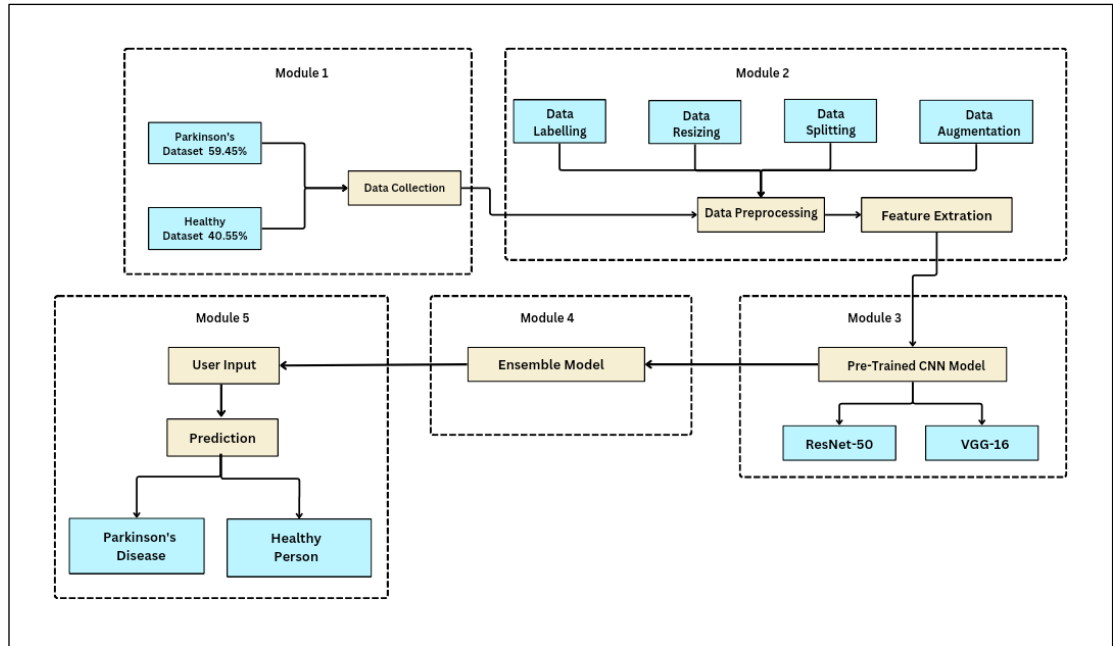


Fig.1 System Architecture

Input dataset: This is the dataset of brain MRI images that will be used to train and evaluate the deep learning model. The dataset should include images from both PD patients and healthy controls.

Pre-processing of PPMI dataset: This step involves cleaning and preparing the PPMI dataset for use in the deep learning model. This may involve tasks such as DCM to JPG conversion, image resizing, data labelling, data splitting and data augmentation.

Input image: This is the brain MRI image that will be used to make a prediction about whether or not the individual has PD.

CNN models: We are using two pretrained models - ResNet50 and VGG16. **3.5 Filter:** A filter is a small matrix of weights and the convolution operation used for extracting features from the input images.

Convolutional layer: This layer is made up of a number of filters. Its output represents the features extracted from input images.

Pooling: This layer aggregates the values of nearby pixels by minimizing the dimensions of the feature map. It reduces complexity of the model and improves its robustness to noise.

Fully Connected Layer: The neurons in this layer are interconnected with every neuron in the previous layer. The outcome of the fully connected layer is a probability distribution over the PD or healthy control classes.

Category: This is the output of the system, which is the predicted class label for the input image.

4.2 DATA FLOW DIAGRAM

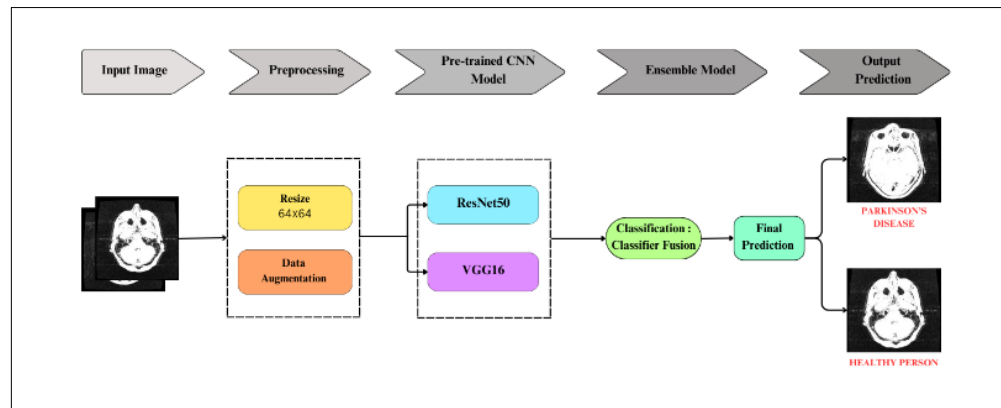


Fig.2 Data Flow Diagram

4.3 COMPONENT DIAGRAM

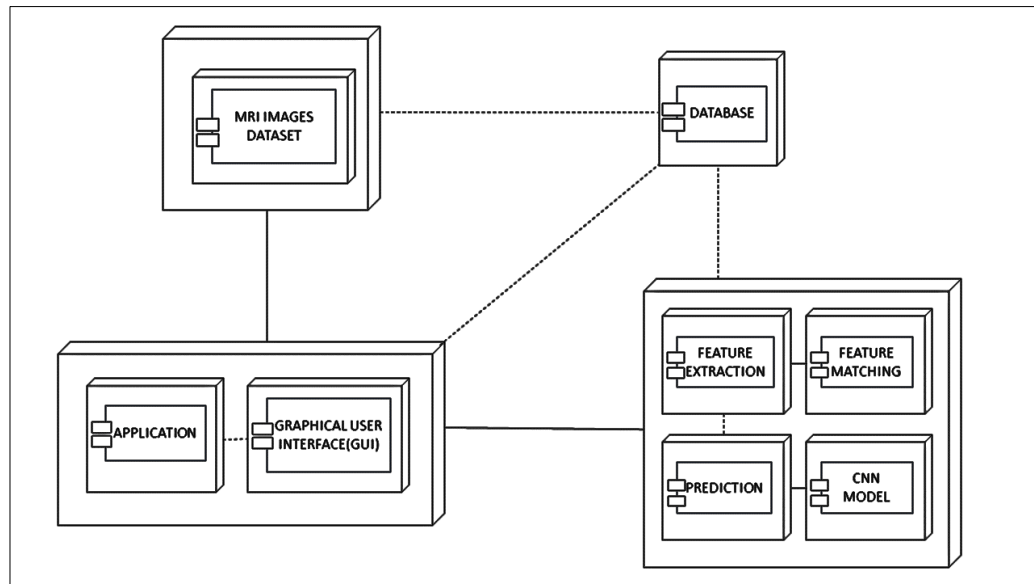


Fig.3 Component Diagram

4.4 CLASS DIAGRAM

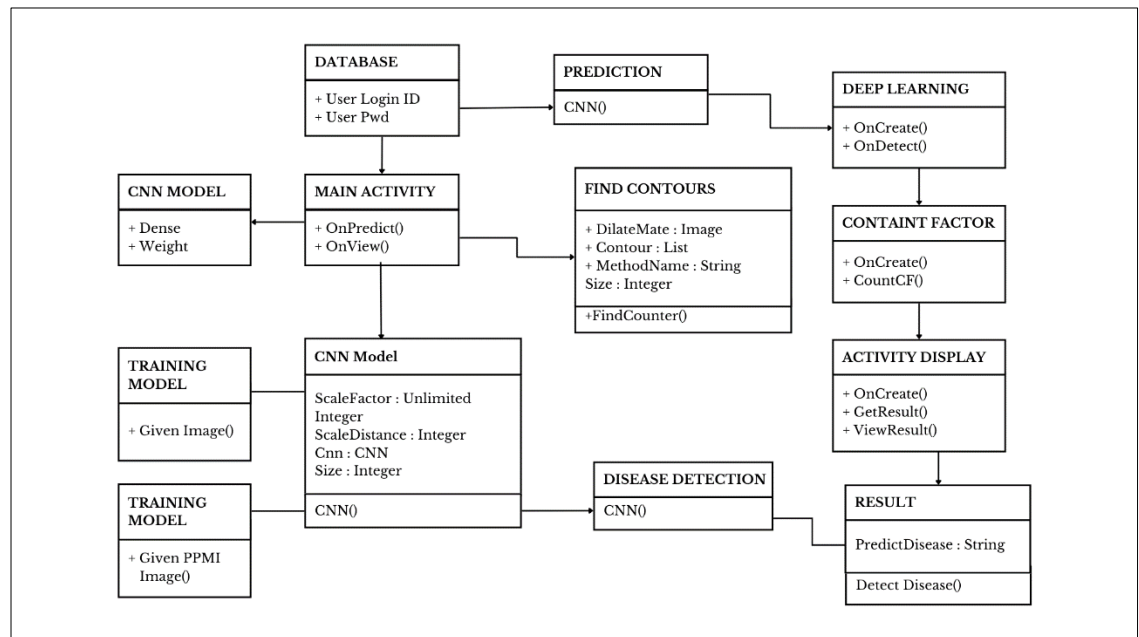


Fig.4 Class Diagram

5. PROJECT PLAN

5.1 PROJECT PLAN

The timeline chart shows overall project development schedule from June to Nov 2023.

Table 2: Project Plan

Months Activities	June'23	July'23	August'23	Sept'23	Oct'23	Nov'23
Literature Reviews	√	√				
Requirement Analysis		√				
Designing		√	√			
Experimental Analysis			√	√	√	
Module wise Implementation				√	√	
Testing and Debugging					√	√
Preparation of Project Report						√

5.2 TEAM ORGANIZATION

The project is planned under the guidance of project guide Prof. Bhagyashri Wankar mam and Project manager team.

A dedicated project log-sheet is maintained to trace communication of the project between team members and project guide. Project updates and issues were discussed with guide.

5.2.1 Team Structure

The team consist of 4 members. The roles and responsibilities are assigned in following way-

Shubham Koshti: Dataset Collection and Model building.

Nikita Kshirsagar: Feature Extraction and GUI designing.

Srushti Bawane: Model building and Testing model.

Amisha Jadhav: Pre-processing dataset and Documentation.

6. PROJECT IMPLEMENTATION

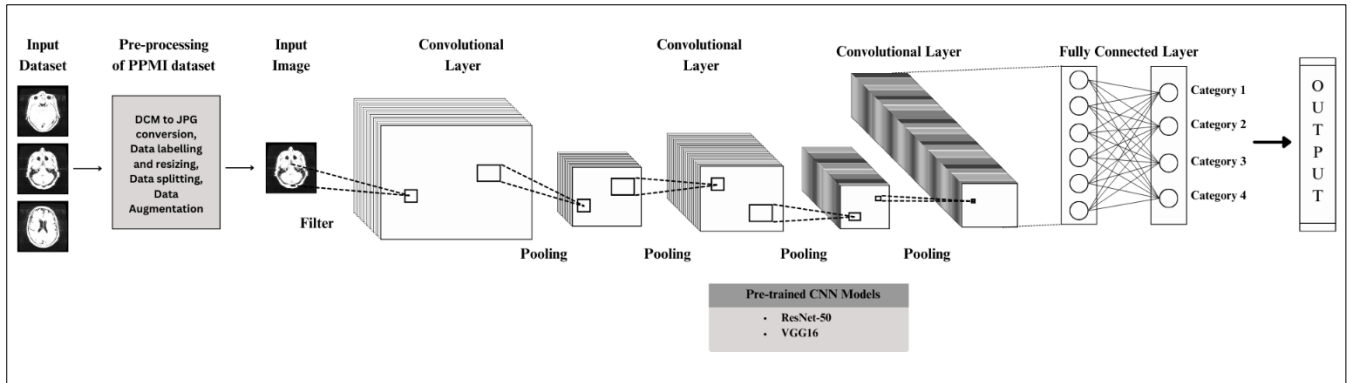


Fig.5 Project Implementation

6.1 OVERVIEW OF PROJECT MODULES

6.1.1 DATASET ACQUISITION

In proposed system, the dataset is gathered from the Parkinson's Progression Markers Initiative. We are Specifically using, the dataset containing a total no of 31,436 MRI scans in the Digital Imaging and Communication in Medicine (DICOM) format. Within this dataset, there are 18,690 MRI scans related to Parkinson's Disease and 12,746 scans representing Healthy Controls.

The splitting of data happened in the following way:

Training: 80%, Validation: 10%, Testing: 10%

Image extension: DICOM (Digital Imaging and Communications in Medicine)

Dataset size: 7.13 GB



Fig.6 Distribution of Classes in the Dataset

6.1.2 PREPROCESSING

Data preprocessing is a crucial stage in which raw data is refined and converted into a more comprehensible format. To prepare this data for our model, we initially converted the DCM images into JPG format and resized them uniformly to dimensions of 64x64 pixels and subsequently applied labeling to the images, ensuring that they were properly annotated and interpretable by our deep learning model. As part of our data preprocessing pipeline, we implemented a data splitting technique. In addition, we employed data augmentation techniques on the training dataset, effectively increasing its size and diversity. This augmentation strategy enabled our model to train on a broader range of images, enhancing its ability to recognize and classify patterns.

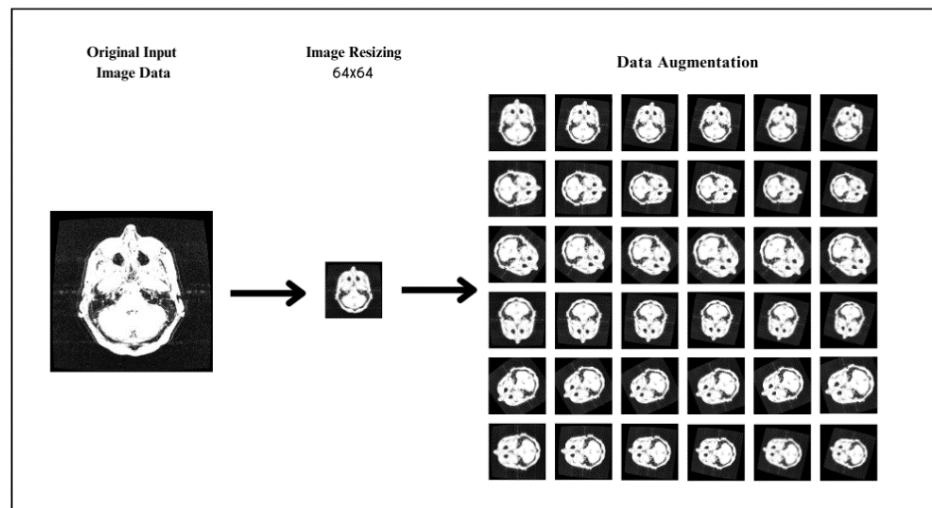


Fig.7 Data Augmentation

6.1.3 MODEL TRAINING

The training of our CNN model for PD detection is guided by a set of crucial parameters and strategies. We maintain the 'weights shuffle' parameter at 'True' to ensure ongoing robust learning. Our 'batch size' remains fixed at 32, facilitating efficient data processing during training. The model continues to refine its weights over a span of 32 training epochs, and we allocate a 'validation split' of 10% to safeguard against overfitting. The input dataset is divided into three proportions: 80% for training, 10% for validation and 10% for testing.

6.1.4 FINE TUNING

Fine-tuning is used for adjusting a pre-trained model for a specific task or dataset, adjusting parameters until the model achieves the desired accuracy. This technique capitalizes on transfer learning, harnessing the model's prior knowledge while adapting it for PD image analysis.

6.1.5 FEATURE EXTRACTION

Feature Extraction, a mathematical-statistical procedure that extracts quantitative parameters of resolution changes/abnormalities from images, will be performed in the initial layers of the CNN architecture using convolutional operations.

6.1.6 TRAINING MODEL

Once the architecture is developed, the model will be trained using the PPMI dataset. TensorFlow and Keras will be used for training.

6.1.7 MODEL EVALUATION

The accuracy of the model will be evaluated with the help of a test dataset in the future.

6.1.8 RETRAINING MODEL

Tuning: The model will be tuned by changing its parameters until the desired accuracy is achieved.

Storing the Weights: The final model will be saved so that it can be used for new data.

6.1.9 WEB-APP DEVELOPMENT

We have used Flask web framework for web app development in python and HTML, CSS, JavaScript for Front end development.

On this website, firstly users need to login/ signup with their accurate credentials. Only after successful login user can upload the image of brain MRI scan. After processing, our website will show the result whether the person has Parkinson's disease or not for the uploaded image.

6.1.10 SERVER-SIDE PROCESSING

Once a user uploads an image, the application will preprocess the image, and the trained model will predict whether a person has Parkinson's disease or not.

6.1.11 CONVOLUTIONAL NEURAL NETWORK (CNN)

CNN is suitable for medical image classification and analysis as they are able to learn complex patterns in images and identify the hidden patterns and trend of data. CNNs consists convolutional layers, pooling layers, and fully connected layers. In the convolutional layer, the features like edges, shapes and textures are extracted by applying the filters on input image. The output of this layer is served as the input for the pooling layers. The working of pooling layer is to keep the important features by reducing the dimensions of the image. Afterwards, to predict and classify the input image, the results from the pooling layers are transmitted to fully connected layers. In our CNN model, the first functional layer is based on VGG16, and the second functional layer is derived from ResNet50. In average layer, an ensemble model often emerges by combining the strengths of ResNet-50 and VGG-16 algorithms. The optimization of model attributes, such as weights and learning rates, is facilitated by the Adam optimizer. To prevent overfitting, we implement the Early Stopping callbacks function from Keras. After training, we save the model so that we can use it later to make predictions in the future.

Input Layer: This layer starts by taking the raw pixel values of an image as its input.

Convolution Layer: Next, it moves to the convolution layer, which extracts important features from the input image. To do this, it uses a filter or kernel along with the image data,

creating something called a feature map that highlights significant patterns.

Pooling Layer: After that, there's a pooling layer that helps simplify things by reducing the amount of data. It picks out the most important values from the feature maps.

Flattening: The data then goes through a process where it's transformed into a 1-dimensional array, which is a simple way of organizing the information.

Fully Connected Layer: This is like a densely connected network that brings all the features together, sort of like assembling pieces of a puzzle to understand the complete picture.

Output Layer: Finally, the output layer combines all the information from the previous layers to produce a single output. This output is used to make a final decision or prediction, like classifying an image.

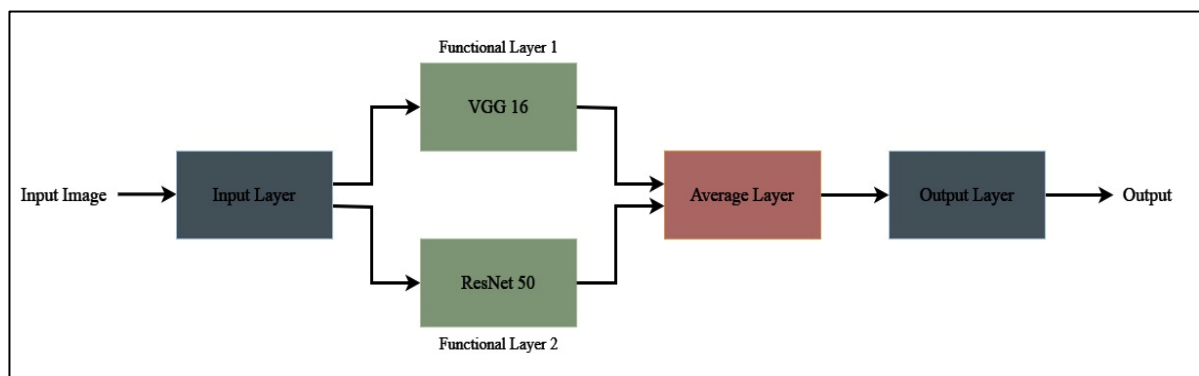


Fig.8 CNN Architecture

6.2 TOOLS AND TECHNOLOGY USED

- Operating System (OS): Windows 7 or higher version
- Integrated Development Environment (IDE): Visual Studio Code
- Programming Language: Python, CSS, HTML, Javascript
- Database: phpMyAdmin (MySQL)
- Libraries: Keras, NumPy, CV2, sklearn, Matplotlib, SciPy, TensorFlow, Flask
- Web Browser: Compatible with any web browser

6.3 ALGORITHM DETAILS

The two widely used CNN algorithms named VGG-16 and ResNet-50 are used to develop PD classifier/model.

6.3.1 ALGORITHM 1

VISUAL GEOMETRY GROUP (VGG-16)

VGG is an acronym for Visual Geometry Group. It is a deep convolutional neural network which was proposed by Karen Simonyan and Andrew Zisserman. The architecture of VGG-16 is shown in figure no 8.

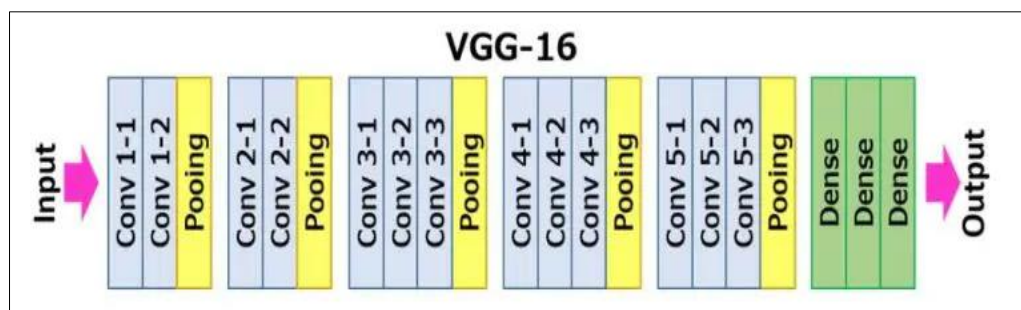


Fig.9 Architecture of VGG-16 model

VGG-16 has 13 convolutional layers, 5 max-pooling layers, and 3 fully connected layers. At initial level, pre-trained ImageNet weights are used later the weights are adjusted to minimize the error rate. The model contains Input layer followed by series of Convolutional layer and Pooling layer. The model is finished by one output layer.

6.3.2 ALGORITHM 2

RESIDUAL NETWORK (ResNet50)

ResNet is an acronym for Residual Networks. It is a classic neural network used as backbone for many computer visions related tasks. The architecture of ResNet-50 is shown in figure no 9.

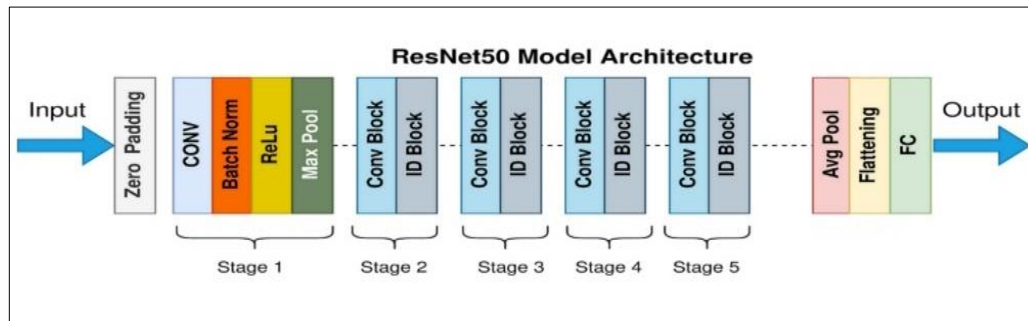


Fig.10 Architecture of ResNet50 model

ResNet50 is one of the variant of ResNet model. The architecture has 5 stages. It has 48 Convolutional layers along with 1 Max Pooling layer and 1 Average Pooling layer. The model is pre-trained by ImageNet weights, which are later adjusted to minimize error rate.

7. RESULT AND DISCUSSION

7.1 OUTCOMES

7.1.1 Accuracy

These models collectively achieved an outstanding accuracy rate of 97%. Leveraging the deep learning capabilities of ResNet-50 and VGG16, we harnessed their unique strengths to extract intricate patterns from medical images, enabling precise identification of early-stage Parkinson's disease. The input image uploading is possible on the system, and the system gives an appropriate output about if the person is diagnosed with PD or not.

Table 3: Comparison Table

Sr No.	Algorithm used	Activation Function	Accuracy
1	VGG-16 ResNet-50 EfficientNetB0	Softmax Sigmoid	0.54 %
2	VGG-16 ResNet-50 EfficientNetB1	Softmax Sigmoid	0.58 %
3	VGG-16 ResNet-50	ReLu Sigmoid	0.97 %

7.1.2 Confusion Matrix

We used confusion matrix for the evaluation of our CNN based prediction model. It determines the model's performance on a test dataset by analyzing the predicted outcomes. The matrix represents the relationship between actual and predicted classes and helps in examining the better performance of model regarding metrics including precision, recall, accuracy, and F1 score.

		Predicted Values	
		Negative	Positive
Actual Values	Negative	32	5
	Positive	0	36

Fig.11 Confusion Matrix

By assessing the of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) distribution, researchers and healthcare professionals can obtain perspectives about model's accuracy and its capacity to reduce misclassifications, which is critical in the context of disease diagnosis.

$$Precision = \frac{TP}{TP+FP} \qquad Recall = \frac{TP}{TP+FN}$$

The F1 score combines these two metrics into a single value, striking a balance among recall and precision. It is calculated as:

$$F1\ score = \frac{2}{\frac{1}{precision} + \frac{1}{Recall}}$$

$$= 2 \times \frac{precision \times Recall}{precision + Recall}$$

The F1 provides a balanced assessment of a model's precision and recall, making it particularly relevant in medical diagnostics like PD detection, where both the terms false negatives and false positives can have significant consequences.

In PD detection with CNNs:

Precision (Positive Predictive Value): This component of the F1 score calculates the accuracy of positive predictions. In the context of PD detection, precision assesses the amount of precisely determined PD cases among the total cases predicted as PD. A higher precision indicates fewer false positive predictions, which is essential for minimizing misdiagnoses.

Recall (Sensitivity): The recall evaluates how well the model can detect all true positive instances. PD detection, it quantifies the ratio of correctly identified PD cases to all the true PD cases. A higher recall implies fewer false negatives, ensuring that actual PD cases are not missed.

Table 4: Classification Report

	Precision	Recall	F1-score	Support
0	1.00	0.86	0.93	37
1	0.88	1.00	0.94	36
Accuracy			0.93	73
Macro avg	0.94	0.93	0.93	73
Weighted avg	0.94	0.93	0.93	73

7.1.3 Visual Graph

We have derived this graph using following algorithms and activation functions:

Algorithm: VGG16, ResNet50 and EfficientNetB0

Activation Function: SoftMax, Sigmoid

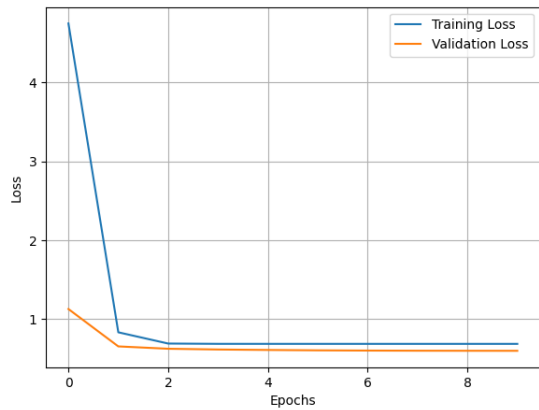


Fig.12 Training vs Validation Loss 1

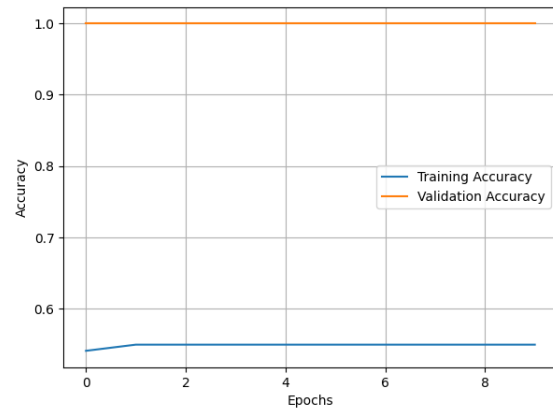


Fig.13 Training vs Validation Accuracy 1

Algorithm: VGG16, ResNet50 and EfficientNetB1

Activation Function: SoftMax, Sigmoid

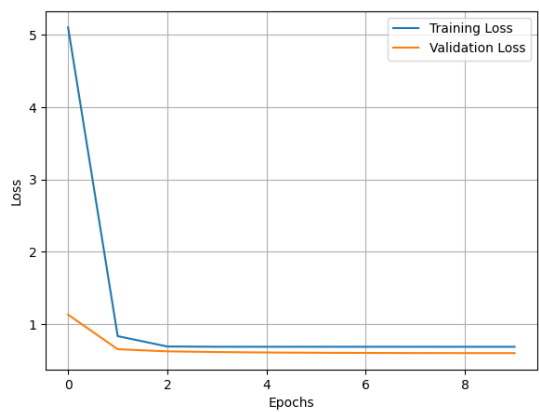


Fig.14 Training vs Validation Loss 2

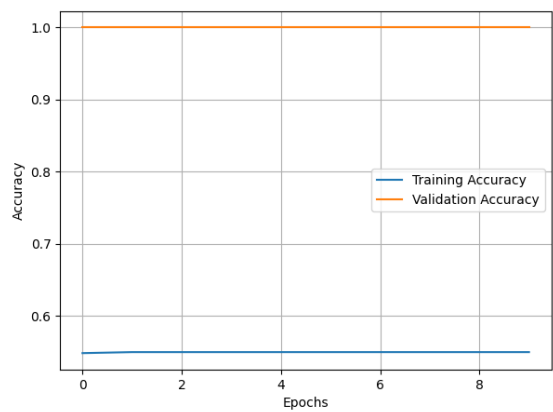


Fig.15 Training vs Validation Accuracy 2

Algorithm: VGG16, ResNet50

Activation Function: ReLu, Sigmoid

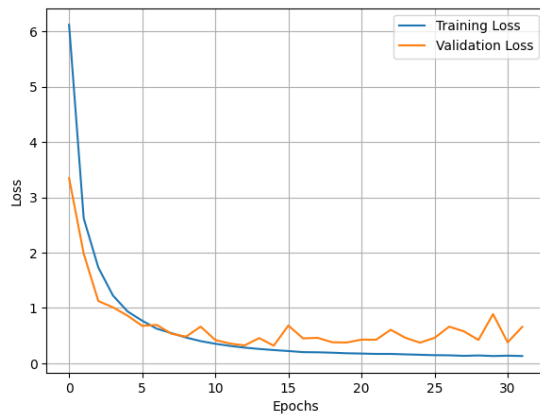


Fig.16 Training vs Validation Loss 3

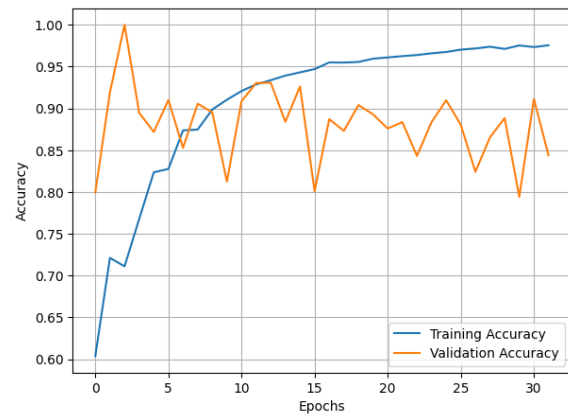


Fig.17 Training vs Validation Accuracy 3

7.2 SCREENSHOTS

7.2.1 Sign Up Page:

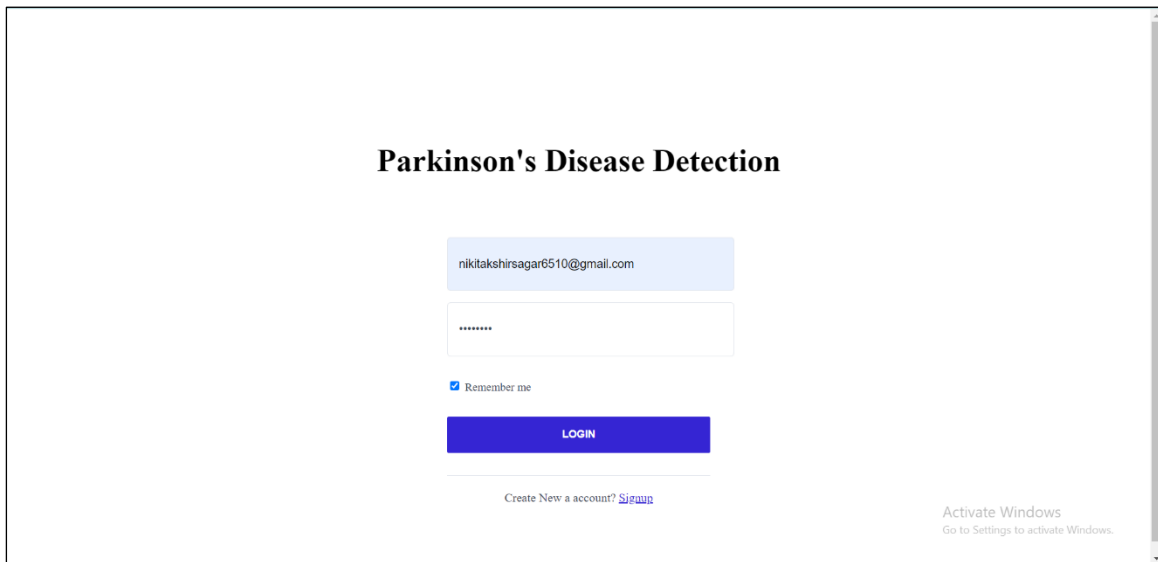
Parkinson's Disease Detection

SIGN UP

Already have an account? [Login](#)

Activate Windows
Go to Settings to activate Windows.

7.2.2 Login Page:



The screenshot shows a login page for 'Parkinson's Disease Detection'. It features a title, a text input field with the email 'nikitakshirsagar6510@gmail.com', a password input field with masked characters, a 'Remember me' checkbox, a blue 'LOGIN' button, and a link to 'Create New a account? Signup'. A Windows activation watermark is visible in the bottom right corner.

Parkinson's Disease Detection

nikitakshirsagar6510@gmail.com

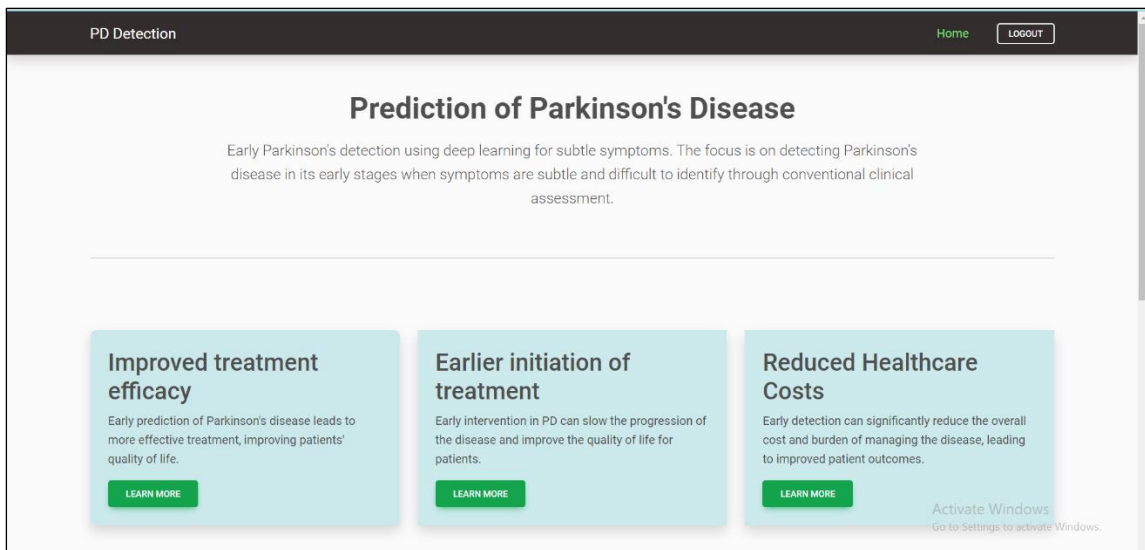
☒ Remember me

LOGIN

Create New a account? [Signup](#)

Activate Windows
Go to Settings to activate Windows.

7.2.3 Home Page:



The screenshot shows the home page of the 'PD Detection' application. It has a dark header with 'PD Detection', 'Home', and 'LOGOUT' links. The main content area is titled 'Prediction of Parkinson's Disease' and includes a descriptive paragraph. Below this are three light blue boxes with titles and descriptions: 'Improved treatment efficacy', 'Earlier initiation of treatment', and 'Reduced Healthcare Costs'. Each box has a 'LEARN MORE' button. A Windows activation watermark is visible in the bottom right corner.

PD Detection Home LOGOUT

Prediction of Parkinson's Disease

Early Parkinson's detection using deep learning for subtle symptoms. The focus is on detecting Parkinson's disease in its early stages when symptoms are subtle and difficult to identify through conventional clinical assessment.

Improved treatment efficacy
Early prediction of Parkinson's disease leads to more effective treatment, improving patients' quality of life.
LEARN MORE

Earlier initiation of treatment
Early intervention in PD can slow the progression of the disease and improve the quality of life for patients.
LEARN MORE

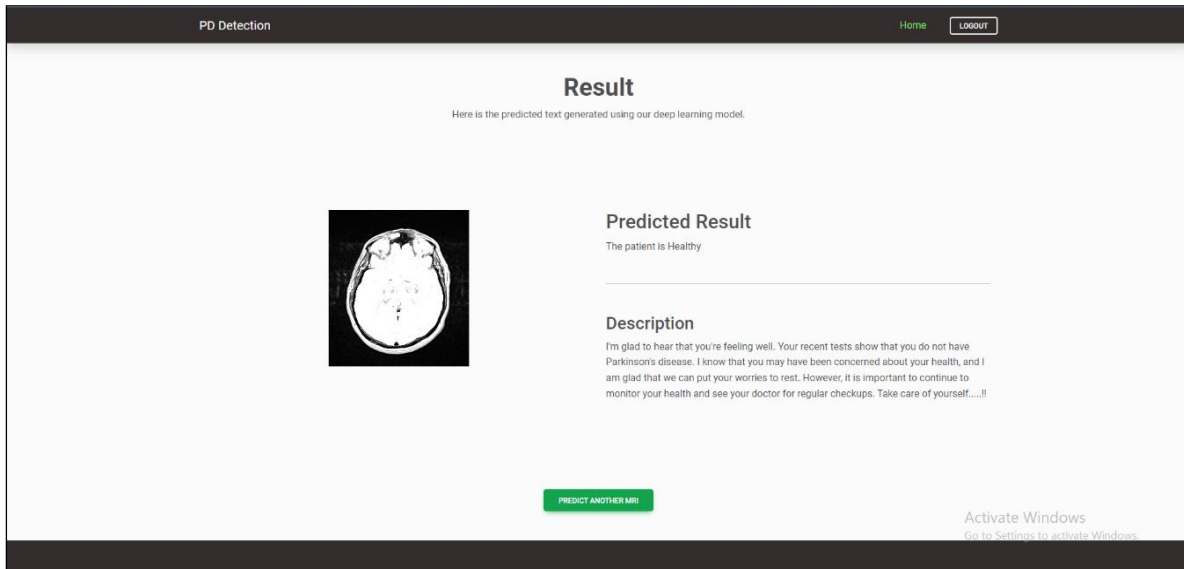
Reduced Healthcare Costs
Early detection can significantly reduce the overall cost and burden of managing the disease, leading to improved patient outcomes.
LEARN MORE

Activate Windows
Go to Settings to activate Windows.

7.2.4 Upload Image of MRI Scan:

7.2.5 Result for Parkinson's Disease:

7.2.6 Result for Healthy:



8. CONCLUSION

8.1 CONCLUSION

In this project, we introduce an innovative convolutional neural network (CNN)-based automated diagnostic system that effectively distinguishes PD from healthy controls (HC). We have used the Parkinson Progression Markers Initiative (PPMI) dataset, which provides publicly available benchmarked magnetic resonance imaging (MRI) images for both PD and healthy controls (HC). It contains a total of 31436 records of MRI scans and is categorized into PD (18690) and healthy control (12746). Furthermore, data is divided into training, validation, and testing sets with a ratio of 80%, 10%, and 10%, respectively.

In conclusion, our system demonstrates a remarkable capability to predict Parkinson's illness in its earlier stages, addressing a critical need in medical diagnostics. By fine tuning key parameters such as kernel size, layers, neurons, and epochs, we can optimize the model, mitigating bias and overfitting concerns. This Deep Learning-based approach opens up exciting opportunities for medical image analysis, empowering researchers and medical teams to engage in feature selection and classification. Ultimately, our model holds the potential to revolutionize PD prediction, providing valuable insights for early intervention and improved patient care. The intersection of deep learning and medical imaging promises a brighter future in the quest to combat Parkinson's disease effectively.

8.2 FUTURE WORK

One key future scope is to extend the model's capabilities by incorporating multi-modal data, including not only brain MRI images but also patient clinical data, genetic information, and vocal features. By integrating these diverse data sources, the model can provide a more comprehensive and accurate diagnosis, potentially revolutionizing early PD detection.

Continuous monitoring of disease progression is an intriguing future scope. Extending the model's capabilities to track PD progression over time can assist in personalized treatment planning and optimize patient care.

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APPENDIX I

List of Publications